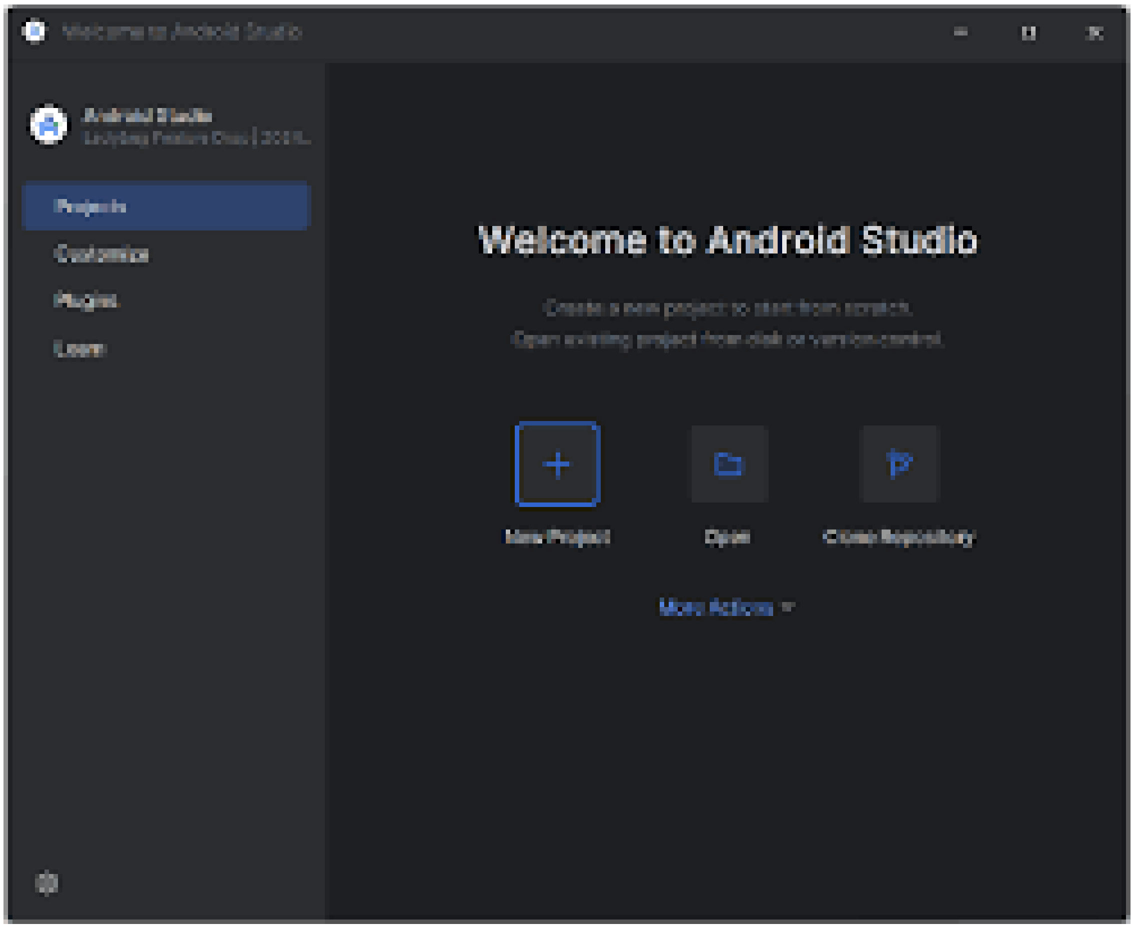
SESSION – 2 :-

Q1:- Install Android Studio on your system, open it, and take a screenshot of the welcome screen showing the version number.



Q2 :- In Android Studio, open the Plugins section and install the Flutter and Dart plugins. Restart the IDE and verify that both plugins are enabled in the settings.

Steps to Install Flutter and Dart Plugins in Android Studio

1. Open Android Studio

Launch Android Studio and wait for it to load.

2. Open Plugins

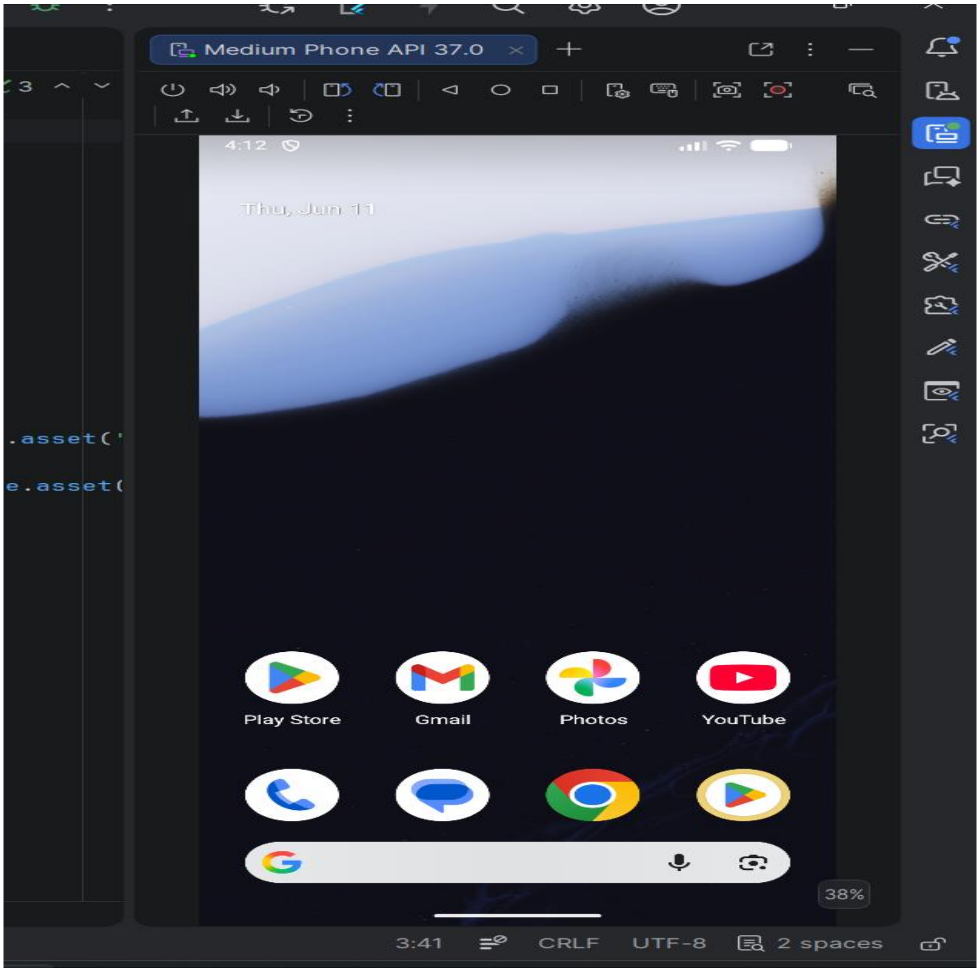
- Click **File → Settings** (Windows/Linux)
- Or **Android Studio → Settings** (macOS)
- Select **Plugins** from the left sidebar

📸 Take a screenshot of the Plugins screen.

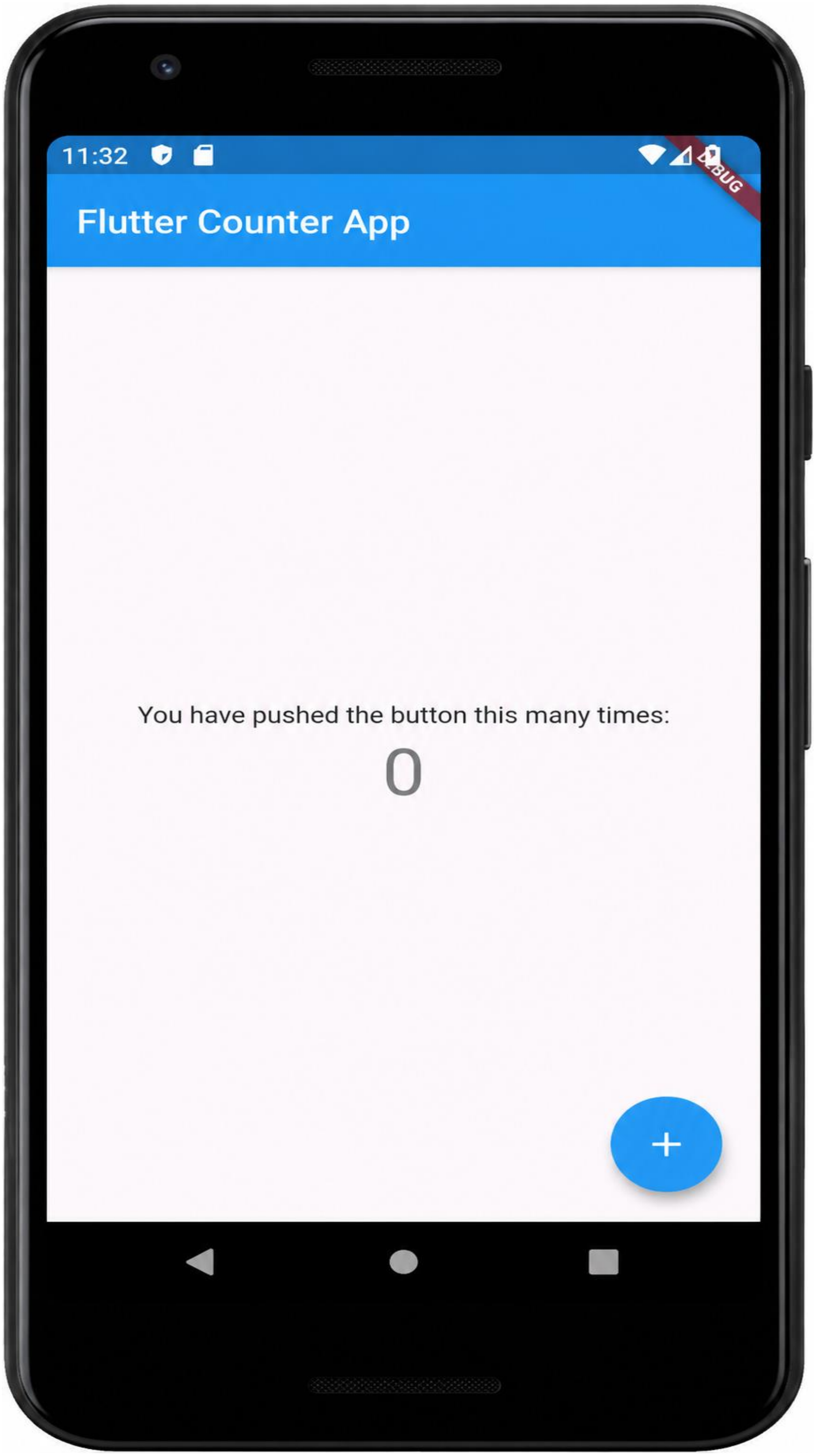
3. Install Flutter Plugin

- Click the **Marketplace** tab
- Search for **Flutter**
- Click **Install**

Q3:-Create a new Android Virtual Device (AVD) in Android Studio with the device name 'Pixel_4_API_30', using any available system image. Start the emulator and share a screenshot of it running.



Q4:-Run the default Flutter counter app on your newly created emulator and confirm that the app launches successfully by sharing the emulator screen with the app running.



Q5:- List three differences you noticed between running your Flutter app on the emulator versus a real Android device. For each difference, explain how it might affect your development or testing process.

Hint: Consider speed, device features, and debugging experience.

Difference	Emulator	Real Android Device	Effect on Development / Testing
Performance & Speed	Runs through your computer's hardware and virtualization, sometimes slower or smoother than actual devices depending on PC specs	Uses real device CPU, GPU, memory, and battery conditions	Performance testing is more reliable on a real device because animations, startup time, scrolling, and responsiveness may behave differently than on the emulator.
Device Features & Hardware Access	Some hardware features are simulated or limited (camera, sensors, GPS, Bluetooth, fingerprint, battery behavior)	Full access to actual hardware components	Features that depend on real hardware should be tested on a physical device to ensure correct behavior and user experience.
Debugging & Development Experience	Easier debugging with quick screenshots, log viewing, hot reload, emulator controls, and reset options	Debugging may take longer due to connecting devices and managing permissions	Emulators speed up UI development and quick testing, while real devices help validate production behavior and catch hardware-specific issues.